

exp: 3

8 Bit Multiplication

AIM:

To write an assembly language program to implement multiplication using 8085 processor.

ALGORITHM:

- 1) Start the program by loading a register pair with the address of memory location.
- 2) Move data to a register.
- 3) Get the second data and load it into the accumulator.
- 4) Add the two register contents.
- 5) Increment the value of carry.
- 6) Check whether repeated addition is over.
- 7) Store the value of product and carry in memory location.
- 8) Halt.

PROGRAM:

MNEMONICS	EXPLANATION
LDA 2200	Load the accumulator with the first number in the address 2200.
MOV E, A	Move the data from accumulator to 'E' register.
MVI D, 00	Move the immediate value 00 into D.

LDA 2201

MOV C, A.

EXI H, 0000

Back: DAD D

DCR C

JNZ BACK

SUBD 2202

HLT

load the accumulator number in the address 2201

move the data from accumulator to 'C'.

load the immediate value 0000 into HL register

Back: label for the loop
DAD D: Add value in D.

decrement register by 1

If C, is not 0, jump to beginning of loop.

store value on HL register pair at memory 2202

HALT.

INPUT:

Address	DATA
2200	4
2201	2

output:

Address	DATA
2202	8

RESULT:

the the program was executed successfully using 8085 processor simulator.



Registers

A	03
BC	00 00
DE	00 05
HL	00 0F
PSW	00 00
PC	42 16
SP	FF FF
Int-Reg	00

Flag

S	0
Z	1
AC	0
P	1
C	0

Load me at

```
1 LDA 2200
2 MOV E,A
3 MVI D,00
4 LDA 2201
5 MOV C,A
6 LXI H,0000
7 BACK: DAD D
8 DCR C
9 JNZ BACK
10 SHLD 2202
11 HLT
12
```

Decimal - Hex Conversion

Decimal

Hex

0	0
To Hex	To Dec

I/O Ports

0	-	00
Update Port Value		

Memory

2201	-	03
Update Memory		

Data Stack KeyPad Memory I/O Ports

Start 2202

OK

Address (Hex)	Address	Data
089A	2202	15
089B	2203	0
089C	2204	0
089D	2205	0
089E	2206	0
089F	2207	0
08A0	2208	0
08A1	2209	0
08A2	2210	0
08A3	2211	0
08A4	2212	0
08A5	2213	0
08A6	2214	0
08A7	2215	0
08A8	2216	0
08A9	2217	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

